

Reporting Your Research Outcomes

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How to write a Research Report?

- Getting ready to write
 - Selection of Sources of related literature
 - Selection of software
 - For creation of Text, Graphics, Image and others
 - Selection of reference styles
 - Selection of Template
- Parts and Order of Manuscript
 - Introduction
 - Motivation
 - Contribution
 - Related Work
 - Background
 - Proposed Method
 - Problem Formulation, Conceptual Framework, Theoretical Foundation, Method Description, Theoretical Analysis
 - Result Section and Discussion
 - Conclusions and Future Work

Getting ready to write

- Selection of Sources of related literature
 - Related Survey Works
 - Their Coverage and Pros and Cons
 - Related Empirical Studies
 - Basis of comparison and selection of winner(s).
 - Other Competing Methods
 - Selection of a set of closed and recent methods for comparison
 - Selection of a set of crucial parameters for comparison
 - How ours is distinct and significant from others?

Getting ready to write

- Selection of software and environment for manuscript creation
 - Text writing software
 - Latex in Windows/Linux with all the packages,
 - MS Word with Mathematics expression writing support
 - Graphics drawing software
 - Power Point and Paint
 - GNU plot
 - Image creation and enhancement software
 - Adobe Photoshop Editor
 - Other packages
 - Audio-video creation software
 - MS Windows Package or X-Windows based or others

Getting ready to write

- MLA Reference Style
 - Borah, B., and D. K. Bhattacharyya. "An improved sampling-based DBSCAN for large spatial databases." *Intelligent Sensing and Information Processing, 2004. Proceedings of International Conference on*. IEEE, 2004.
- APA Reference Style
 - Borah, B., & Bhattacharyya, D. K. (2004). An improved sampling-based DBSCAN for large spatial databases. In *Intelligent Sensing and Information Processing, 2004. Proceedings of International Conference on* (pp. 92-96). IEEE.
- Chicago Reference Style
 - Borah, B., and D. K. Bhattacharyya. "An improved sampling-based DBSCAN for large spatial databases." In *Intelligent Sensing and Information Processing, 2004. Proceedings of International Conference on*, pp. 92-96. IEEE, 2004.
- IEEE Reference Style
 - Borah and Bhattacharyya [3], in their work on improved sampling-based DBSCAN for large spatial databases.....
- Springer Reference Style
 - Borah and Bhattacharyya [3], in their work on improved sampling-based DBSCAN for large spatial databases.....
- Elsevier Reference Style
 - Clustering is a popular analysis tool in data mining applications ([Han and Kamber, 2006](#) and [Roy and Bhattacharyya, 2010](#)) such as scientific data exploration, information retrieval and text mining,
- ACM Reference Style
 - ... So these old single source attacks are now countered easily by many defense mechanisms [7, 13, 1, 3] and source of
- Willey Reference Style
 - As a result, a few studies can be found in the literature attending this phase of the rule mining process.¹⁻³

Getting ready to write

- What are SCI Journals?

- The **Science Citation Index (SCI)** is a [citation index](#) originally produced by the [Institute for Scientific Information \(ISI\)](#) and created by [Eugene Garfield](#). It was officially launched in 1964. It is now owned by [Thomson Reuters](#).^{[1][2][3][4]} The larger version (**Science Citation Index Expanded**) covers more than 6,500 notable and significant [journals](#), across 150 disciplines, from 1900 to the present. These are alternately described as the world's leading journals of [science](#) and [technology](#), because of a rigorous selection process.^{[5] [6][7]}
- The index is made available online through different platforms, such as the [Web of Science](#)^{[8][9]} and SciSearch.^[10] (There are also CD and printed editions, covering a smaller number of journals). This database allows a researcher to identify which later articles have cited any particular earlier article, or have cited the articles of any particular author, or have been cited most frequently. Thomson Reuters also markets several subsets of this database, termed "Specialty Citation Indexes",^[11] such as the **Neuroscience Citation Index**^[12] and the **Chemistry Citation Index**.^[13]
- As per UGC norms, any journal with impact factor (Thomson-Reuter) is considered as SCI journal

- Journal Template

- IEEE Journal /Conference Template
- Springer Journal/Conference Template
- Elsevier Journal/Conference Template
- ACM Journal/Conference Template
- Willey Journal/Conference Template
- Other reputed Journal/Conference Template

What is h index?

- The h -index is a number intended to represent both the productivity and the impact of a particular scientist or scholar, or a group of scientists or scholars (such as a departmental or research group).
- The h -index is calculated by counting the number of publications for which an author has been cited by other authors at least that same number of times. For instance, an h -index of 17 means that the scientist has published at least 17 papers that have **each** been cited **at least** 17 times. If the scientist's 18th most cited publication was cited only 10 times, the h -index would remain at 17. If the scientist's 18th most cited publication was cited 18 or more times, the h -index would rise to 18.
- Part of the purpose of the h -index is to eliminate outlier publications that might give a skewed picture of a scientist's impact. For instance, if a scientist published one paper many years ago that was cited 9,374 times, but has since only published papers that have been cited 2 or 3 times each, a straight citation count for that scientist could make it seem that his or her long-term career work was very significant. The h -index, however, would be much lower, signifying that the scientist's overall body of work was not necessarily as significant.
- The following resources will calculate an h -index:
 - [Scopus](#)
 - [Web of Science](#)
 - [Google Scholar](#)
 - [Pure](#) (MD Anderson Faculty and Fellows listed)

Parts of a Technical Manuscript

- Parts in Order
 - Introduction
 - Motivation
 - Contribution
 - Related Work
 - Background
 - Proposed Method
 - Problem Formulation, Conceptual Framework, Theoretical Foundation, Method Description, Theoretical Analysis
 - Result Section and Discussion
 - Conclusions and Future Work

Parts of a Technical Manuscript

- **Title Selection**
- Authors name, affiliation and sequence
- Author note
 - APA style journal article also includes an author note to express disclaimer, conflict of interest, acknowledgement, etc
- **Abstract**
 - Concisely : problem, importance, contribution, findings, conclusion
 - Keywords
 - Abstract formatting
- **Introduction**
 - What is the issue?
 - Why important?
 - Existing approaches
 - Brief analysis and Motivation
 - What is proposed?
 - Why is it significant?
 - Organization of the manuscript.

Parts of a Technical Manuscript

Contributions

- State your contribution 1
- State your contribution 2, etc

Background

- Basic concepts/techniques, their definitions, expressions and uses
- Selection of concepts/techniques and justifications
- Importance in subsequent section(s).

Related Work

- Theory
- Approaches
- Taxonomy
- Discussion
- Pros and Cons

Problem Formulation

- State the problem in hand

Proposed Method

- Description of the approach introduced for solving the problem
- Symbol/notation table for consistent description and the approach/method
- Framework at conceptual level (use figure) and Its Description
- Theory : Definitions, Propositions, Lemma, Theorems, Conjectures, etc (if any)
- Presentation in Pseudo-code/Algorithm/flowchart or any other form (if relevant)
- Description of the algorithm/Code/flowchart w.r.t. Definitions and other theoretical evidences
- Complexity (how efficient theoretically?)
- Theory
 - Proofs of correctness
 - Proof of completeness
 - Proof of Convergence

Parts of a Technical Manuscript

Results & Analysis

- Environment Design; Selection of software/hardware/equipment
- Dataset design/generation for simulation study
- Selection of real-life or benchmark datasets
- Conditions and Parameter Specifications
- Results finding
 - Simulation results;
 - Experimental Study on Benchmark / real-life data;

Validation & Analysis

- Internal Validation
 - Using Statistical Measures; Cross Validation; Analysis
- External Validation
 - Simulations; Cross Validation; Analysis
 - Using Benchmark/Standards ; Cross Validation; Analysis

Comparison with Prior Works Using Graphical Aids/Tools

- Execution time comparison (e.g., Graph for Data vs Time for each method)
- Accuracy Comparison (e.g., in terms of ROC)
- Comparison in terms of other statistical measures (e.g., Precision, Recall, F measure, t-test, etc)
- Comparison in terms of Scalability (e.g., Graph for Increasing Data vs Time)

Issues and Challenges

- List of Research Issues
- List of Challenging tasks

Conclusion and Future Works

- Concluding remarks: Summary of findings, salient features, etc
- Future Works: Future directions to carry out further research